

Displacement during the Great Recession: Local Labor Markets, Outside Options and Worker Adjustments

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Abstract

This paper examines how regional labor market conditions shape workers' adjustment to job displacement, with a focus on the role of local opportunities and spatial frictions. Using linked employer–employee data from five Great Lakes states during the Great Recession, we analyze the employment and earnings trajectories of workers displaced by mass layoffs. We introduce a new measure of labor market opportunity—the Job Opportunities Index (JOI)—which captures both the quantity of available jobs and their associated earnings, providing a more comprehensive measure of local economic opportunity than standard indicators such as unemployment rates. We find substantial heterogeneity in post-displacement outcomes across labor markets. Workers located in areas with high job opportunities and low unemployment experience significantly smaller earnings losses—approximately half those of workers in less favorable markets. Despite these differences, we find limited evidence that geographic mobility improves earnings trajectories. To understand this pattern, we examine the role of spatial frictions and show that housing costs in nearby labor markets constrain access to higher-opportunity regions. These findings highlight the importance of local opportunity structures and mobility barriers in shaping adjustment to economic shocks, and suggest limits to the equalizing role of worker mobility across regions.

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1 Introduction

The widespread and devastating long-term impacts of the Great Recession on American workers have been well documented (Finkelstein et al., 2024; Rinz, 2022; Rothstein, 2023; Yagan, 2019), with particularly detrimental impacts among workers in the Midwest as well as those in manufacturing (Charles et al., 2019). However, the role of regional labor market conditions in shaping workers’ adjustment to large negative shocks is much less understood. This question is central to regional science, where spatial differences in economic opportunities and mobility frictions play a key theoretical role in determining economic outcomes (Marshall, 1890).

This paper examines how features of local labor markets and outside options jointly shape the employment and earnings trajectories of workers displaced by mass layoffs. We focus on three core dimensions of regional labor markets: unemployment rates, housing costs, and a new measure of job opportunities that captures both the quantity of jobs and their associated earnings. Our central question is how these local conditions and spatial alternatives affect both earnings recovery and the joint decision of where and when to return to work.

We make three main contributions. First, we introduce a new measure of local labor market opportunity, the Job Opportunities Index (JOI), which combines job availability and expected earnings into a single metric. This index complements standard indicators such as unemployment rates and captures an additional dimension of spatial variation in economic opportunity. Second, we document substantial heterogeneity in post-displacement outcomes across labor markets, showing that workers in high job opportunity, low unemployment areas experience significantly smaller earnings losses. In particular, workers in the top decile of the JOI distribution experience no long-run earnings loss on average, relative to non displaced workers. Third, we provide new evidence on the role of spatial frictions in shaping adjustment to job loss. Despite substantial differences in local opportunities, we find limited evidence that geographic mobility improves long-run earnings outcomes, and we show that housing costs in nearby labor markets act as an important barrier to accessing higher-opportunity regions.

To carry out this analysis, we construct a location-specific, worker-level longitudinal dataset by linking near-universal matched employer-employee data from the Longitudinal Employer-Household Dynamics (LEHD) program with measures of local labor market conditions. These data allow us to track workers’ employment histories, earnings, and geographic mobility for five years following displacement, as well as to observe the mass layoff events that initiate jobless spells.

Our empirical approach builds on the literature estimating earnings losses following displacement (Jacob-

son et al., 1993; Couch and Placzek, 2010; Davis and von Wachter, 2011; Lachowska et al., 2020; Schmieder et al., 2023). We estimate an earnings model with worker fixed effects and dynamic treatment effects, allowing earnings paths to vary with initial labor market conditions and industry. A key feature of our approach is the construction of two comparison groups. In contrast to the standard practice of pooling all non-displaced workers within a labor market, we distinguish between workers who are employed at a mass-layoff establishment and those employed at non-mass-layoff establishments in the same MSA. We show that workers in mass layoff establishments who are not displaced experience substantial earnings losses, and we therefore focus on workers in non-mass-layoff establishments as our primary comparison group.

Our earnings analysis relates to a growing literature on outside options and labor market dynamics. Caldwell and Danieli (2024) show that better outside options are associated with higher wages, while Olivares (2023) finds that nonlocal labor demand influences wage growth. We contribute to this literature by examining how both local conditions and spatial alternatives shape the recovery paths of displaced workers. Our work is also closely related to studies of labor market thickness and spatial mismatch. For example, Moretti and Yi (2024) show that workers displaced from firm closures experience better outcomes in larger labor markets, while Haller and Heuermann (2020) highlight the tradeoff between job opportunities and competition. Rather than focusing on a single dimension such as market size or density, we consider multiple dimensions of labor market conditions and explicitly incorporate outside options.

In addition to earnings dynamics, we examine the duration and location of re-employment using a competing risks framework. Motivated by evidence that non-employment duration plays a central role in shaping earnings losses (Fallick et al., 2025), we model the probability of exiting joblessness to either the origin MSA or an alternative destination. We incorporate both local and outside labor market conditions, including unemployment rates, job opportunities, and housing costs, and allow for unobserved heterogeneity following Heckman and Singer (1982).

Our results highlight the importance of regional labor market conditions in shaping a worker’s adjustment to displacement. Differences in local opportunity structures translate into large differences in earnings recovery, yet these disparities are not fully offset by worker mobility. Instead, our findings point to an important role for spatial frictions: housing costs in outside option labor markets limit access to higher-opportunity regions, constraining the extent to which workers can improve outcomes through relocation. These patterns suggest that the benefits of stronger labor markets are not equally accessible, even within relatively integrated regional systems.

The Great Recession and the concentration of job loss in the Midwest provide a particularly informative setting in which to study these mechanisms. The severity of the shock and the pronounced spatial variation in industrial composition and labor market conditions generate substantial cross-market differences in

opportunities, allowing us to more clearly identify the role of local conditions and mobility constraints in shaping worker outcomes.

Taken together, these findings underscore the distinction between the availability of economic opportunities and workers’ ability to access them. While spatial variation in labor market conditions creates the potential for improved outcomes, mobility barriers limit the equalizing role of migration, contributing to the persistence of regional disparities following large economic shocks.

The remainder of the paper proceeds as follows. Section 2 provides a conceptual framework for our analysis. Section 3 embeds our analysis in existing literature. Section 4 describes the data. Section 5 provides summary statistics. Section 6 outlines the empirical framework. Section 7 presents the results, and Section 8 concludes.

2 Conceptual Framework

Our analysis examines how local labor market conditions and outside options shape the adjustment of displaced workers following a mass layoff. We focus on three key outcomes: the duration of non-employment, the location of re-employment, and subsequent earnings. These outcomes are jointly determined by the availability of economic opportunities across space and by the frictions that limit workers’ ability to access those opportunities.

We begin from a standard job search framework in which displaced workers receive job offers drawn from a wage offer distribution and accept employment when an offer exceeds their reservation wage. In this setting, both the arrival rate of job offers and the distribution of wages depend on labor market conditions. We extend this framework to a spatial setting in which workers receive offers not only from their origin labor market but also from nearby labor markets that constitute outside options. This extension reflects a long tradition in regional economics emphasizing the role of agglomeration and market thickness in shaping labor market outcomes (Marshall, 1890; Neffke et al., 2018).

In our framework, a worker displaced in period $t = 1$ in metropolitan area m chooses among three possible states in each subsequent period: non-employment, employment in the origin MSA, or employment in an outside MSA. Let $L_{it} \in \{0, 1, 2\}$ denote this outcome. The worker’s indirect utility depends on individual characteristics, local labor market conditions, and the characteristics of outside options:

$$U_{it} = f(X_{it}, MSA_{mt}, MSA_{mt}^N) \quad (1)$$

where X_{it} includes worker characteristics such as age, prior earnings, tenure, industry, race, and gender;

MSA_{mt} captures conditions in the origin labor market; and MSA_{mt}^N summarizes conditions in nearby labor markets that form the worker’s outside option.

We operationalize local and outside labor market conditions using three measures: the Job Opportunities Index (JOI), the unemployment rate (UNEMP), and the housing cost index (HCI). The JOI captures both the quantity of available jobs and their associated earnings, providing a measure of economic opportunity. The unemployment rate reflects labor market tightness and job availability, while housing costs capture a key spatial friction affecting the feasibility of relocation. For each worker, outside options are defined as a weighted average of characteristics across the five most likely destination MSAs, based on observed job-to-job flows.

This framework highlights two key channels. First, differences in local labor market conditions affect both the probability of receiving job offers and the quality of those offers, shaping earnings trajectories after displacement. Second, spatial frictions—captured here by housing costs—affect the extent to which workers can access opportunities in other labor markets. As a result, the availability of opportunities and the ability to access them are distinct, and both play a role in determining outcomes.

We examine two dimensions of post-displacement adjustment. The first is the path of earnings following displacement, which has been the focus of a large literature documenting persistent losses (Jacobson et al., 1993; Couch and Placzek, 2010; Davis and von Wachter, 2011; Lachowska et al., 2020; Schmieder et al., 2023). The second is the duration of non-employment and the location of re-employment. Motivated by recent work showing that non-employment duration is a key determinant of long-run earnings losses (Fallick et al., 2025), we model the transition from non-employment to employment as a competing risks problem, with exits to either the origin or an outside labor market.

This approach allows us to link local labor market conditions and spatial frictions to both the timing and location of re-employment, and, through these channels, to longer-run earnings outcomes. In doing so, we connect the literature on job displacement and earnings losses with work in regional science on agglomeration, labor market thickness, and spatial mobility.

3 Literature and Context

Our paper contributes to three related strands of literature on the impacts of economic shocks, the recovery of displaced workers, and the role of local labor market conditions.

First, a large body of work documents the long-term impacts of the Great Recession on workers. Foote et al. (2019) examine how local labor markets adjust to mass layoffs, showing that while out-migration historically accounts for a large share of labor force adjustment, non-participation played a more prominent

role during and after the Great Recession. At the individual level, [Song and von Wachter \(2014\)](#) and [Yagan \(2019\)](#) provide evidence of persistent declines in employment following the Great Recession, with particularly large effects among lower-income workers. Using linked employer-employee data, [Abowd et al. \(2018\)](#) highlight the importance of employer heterogeneity in shaping earnings outcomes. Our paper contributes to this literature by examining how regional labor market conditions mediate the long-run impacts of displacement during this period.

Second, our work relates to a literature examining re-employment dynamics and the duration of joblessness following displacement. [Andersson et al. \(2018\)](#) show that better job accessibility within a metropolitan area reduces the duration of joblessness, while [Hellerstein et al. \(2019\)](#) find that stronger residence-based labor market networks facilitate re-employment. Bridging these perspectives, [Fallick et al. \(2025\)](#) demonstrate that the length of a non-employment spell is a key determinant of long-run earnings losses. We build on this work by examining how both local and outside labor market conditions jointly shape the duration of non-employment and, through this channel, earnings recovery.

Third, our paper contributes to a growing literature on how local labor market characteristics influence the recovery of displaced workers. [Bleakley and Lin \(2012\)](#) document lower labor market churn in denser areas, including among displaced workers. [Neffke et al. \(2018\)](#) show that local industry composition affects re-employment prospects and earnings losses, while [Kostea \(2019\)](#) highlight the importance of occupational distance relative to local labor markets. More recently, [Macaluso \(2025\)](#) demonstrates that workers in skill-remote jobs experience worse re-employment and earnings outcomes. Closely related, [Moretti and Yi \(2024\)](#) show that labor market size improves re-employment prospects following firm closures. Our work complements this literature by focusing on broader measures of local economic opportunity and explicitly incorporating outside options and spatial frictions.

Taken together, these strands of literature highlight the importance of both local labor market conditions and re-employment dynamics in shaping the consequences of displacement. Our contribution is to integrate these perspectives in a spatial framework that distinguishes between the availability of opportunities and workers' ability to access them.

4 Data and Sample Construction

The primary data used in this paper are drawn from the U.S. Census Bureau's confidential Longitudinal Employer-Household Dynamics (LEHD) dataset. The LEHD is a matched employer-employee panel that follows most U.S. workers over time, covering more than 150 million private-sector employees and, beginning in 2011, state and local government employees. The data are constructed from multiple administrative

sources, with the core input coming from state Unemployment Insurance (UI) earnings records. For most states, the data begin in 1999 and provide quarterly information on workers’ employment, earnings, industry, and location, along with demographic characteristics such as race, gender, county of birth, and imputed education. The LEHD has been widely used to study worker mobility and labor market dynamics (Abowd et al., 2009; Andersson et al., 2018; Pollakowski et al., 2022; Moretti and Yi, 2024).

Our analysis focuses on displaced workers in five Great Lakes states: Indiana, Wisconsin, Ohio, Pennsylvania (excluding Philadelphia), and New York (excluding New York City). These states are drawn from the set of 28 states for which we have full access to detailed LEHD data. We restrict attention to prime-age workers (ages 25–55) who earned at least \$15,000 in the year prior to displacement and who were employed at the same establishment for this full time period. This restriction allows us to focus on more attached workers, for whom job loss is more likely to reflect firm-level shocks rather than individual circumstances. The LEHD allows us to observe workers’ earnings and employment histories within our sample states, including workers who move across these states. For moves to states outside our sample, we observe the timing of re-employment but not post-move earnings (Vilhuber, 2018).

We construct two samples of displaced workers. The first sample includes 143,000 workers displaced during the Great Recession (2007 Q4–2009 Q4) for whom we observe complete post-displacement earnings histories, either because they remain in their original state or move to another state within our sample. This sample is used to estimate earnings trajectories following displacement. The second sample expands the analysis to include workers who move to states outside the set of participating states, increasing the sample to 162,000 displaced workers. Because earnings are not observed for these workers after re-employment, this sample is used to analyze non-employment duration and re-employment patterns. In this context, we define re-employment in a stable job as four consecutive quarters of employment.

4.1 Local Labor Market Measures

Our analysis focuses on three dimensions of local labor market conditions that theory suggests are central to displaced workers’ adjustment: job opportunities, labor market tightness, and housing costs. We capture these dimensions using the Job Opportunities Index (JOI), the unemployment rate (UNEMP), and a housing cost index (HCI), respectively.

The JOI is designed to measure the expected employment value of a labor market, combining both the likelihood of obtaining a job and the expected earnings conditional on employment. Conceptually, it reflects the quantity and quality of job opportunities available to workers. A key feature of the JOI is that it is constructed separately by education group, allowing for heterogeneity in opportunities across skill levels. We

use education information from the LEHD, which is imputed by the Census Bureau and categorized into three groups: high school or less, some college, and a bachelor’s degree or more. Because the JOI is constructed separately by education group, part of its variation reflects differences in labor market opportunities across skill levels, which we account for in our empirical analysis.

To construct the JOI, we use aggregate Job-to-Job (J2J) flows from the LEHD. For each MSA, education group, and quarter, the JOI is defined as the product of average earnings among recent job switchers into that MSA and a measure of hiring intensity, defined as hires relative to total employment. Intuitively, this measure captures the total earnings potential available to workers through new job opportunities.

Among alternative measures of hiring intensity, we focus on the net hiring rate, which accounts for both job creation and separations and therefore better reflects the availability of stable employment opportunities. As shown in Appendix A, which provides a more detailed explanation of how this index is constructed, this measure is more strongly associated with earnings gains for job switchers than alternative specifications.

The JOI complements more standard measures of labor market conditions such as the unemployment rate, which captures labor market slack but does not reflect the earnings potential of available jobs (Schmieder and von Wachter, 2010; Valletta, 2013). While related to measures such as vacancies or quits (Heise et al., 2026), the JOI combines information on both hiring and wages, providing a more comprehensive measure of local opportunity. Empirically, the JOI and unemployment rate are strongly negatively correlated, with a median within-MSA correlation of -0.79 and a correlation of -0.94 across MSAs after controlling for MSA and time fixed effects. Despite this high correlation, we show below that both measures retain predictive power for post-displacement outcomes, suggesting they capture distinct aspects of local labor markets.

In addition to local conditions, we construct measures of labor market conditions in workers’ outside options. For each origin MSA, we identify the five most common destination MSAs based on job-to-job flows and define the outside option as a weighted average of these destinations, with weights given by observed worker flows. This approach captures the most relevant alternative labor markets available to displaced workers. All else equal, stronger labor market conditions in either the origin or outside option are expected to improve re-employment prospects.

Finally, we measure housing costs using MSA-level house price indices. We begin with a baseline measure of house prices estimated from the 2000 Decennial Census using a hedonic pricing model, and update this measure quarterly using indices from the Federal Housing Finance Agency (FHFA). Housing costs capture an important spatial friction: higher costs in alternative labor markets may limit workers’ ability to relocate and access better opportunities. We therefore expect housing costs—particularly in outside option markets—to affect both the duration of non-employment and the location of re-employment.

4.2 Defining Mass Layoffs

We focus on mass layoffs during the Great Recession because these separations are especially likely to reflect changes in economic conditions rather than idiosyncratic worker-level factors. Relative to ordinary separations, mass layoffs provide a clearer measure of involuntary displacement and are therefore well suited to our analysis of how local labor market conditions shape post-displacement outcomes.

We define displaced workers as those who lose their jobs due to a mass layoff, including establishment closures. We identify mass layoffs at the establishment level rather than the firm level. This choice is important because multi-establishment firms may concentrate employment reductions in particular locations rather than spread layoffs evenly across the firm. Defining layoffs at the establishment level therefore better captures targeted local employment shocks. It is also more practical in our setting, since some firms operate across multiple states while our detailed worker-level earnings data are available only for a subset of states. We focus on workers' first observed mass layoff spell, as subsequent layoffs may reflect outcomes that are influenced by earlier displacement events.

Following [Jacobson et al. \(1993\)](#), we define a mass layoff as an employment reduction of at least 30 percent within a four-quarter period at an establishment with more than 50 workers. Following [Hellerstein et al. \(2019\)](#), we further restrict attention to workers with relatively strong prior labor force attachment, requiring at least four quarters of tenure at the establishment and annual earnings of at least \$15,000 prior to the start of the layoff period.¹

We take several steps to ensure that identified events correspond to genuine mass layoffs. First, we require that the event be preceded by four consecutive quarters of establishment stability, defined as either employment growth or employment losses of less than 30 percent. Second, we exclude cases in which workers appear to move collectively to another establishment, as may occur following an ownership change, reorganization, bankruptcy, or buyout. Third, when establishment-level employment records are missing for a given year, we do not classify the resulting disappearance as an establishment closure.

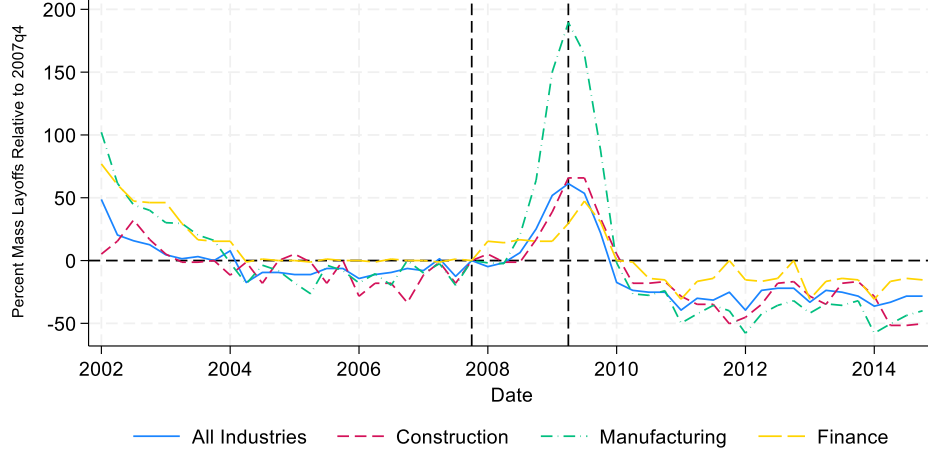
A further complication arises because the LEHD is built from worker-level records reported by states at the firm level. For multi-establishment firms, workers must therefore be probabilistically assigned to establishments. This is not an issue for single-establishment firms, but it introduces additional measurement challenges for larger firms. [Appendix B](#) provides further details on our treatment of these cases.

[Figure 1](#) plots the number of mass layoffs in the five sample states from 2002 to 2015. The figure shows a sharp rise in layoffs during the Great Recession and persistently elevated levels immediately afterward, particularly in manufacturing. This pattern supports our focus on the recession period and our decision to

¹We exclude workers who are rehired by the same establishment within eight quarters of separation, as these cases are less likely to reflect permanent displacement.

extend the mass layoff window two quarters beyond the conventional recession dates.

Figure 1: Mass Layoffs, Percent Difference from 2007 Q4



Notes: Calculations use LEHD data for all establishments in the five sample states, across all industries and within three key sectors. The figure reports estimates of establishment-level mass layoff events from 2002 to 2015. Estimates are seasonally adjusted. Vertical lines indicate the beginning and end of the Great Recession.

One potential concern is that some workers may leave declining establishments before the mass layoff occurs. If such workers differ systematically from those who remain, estimates based on mass layoffs could be affected by selective attrition. To assess this possibility, we construct a sample of “Early Leavers,” defined as workers who exit the establishment in the year prior to the mass layoff, and a sample of “Very Early Leavers,” defined as workers who exit in the year before that. Appendix C, Table C1 shows that early leavers are broadly similar to both very early leavers and displaced workers, suggesting that anticipatory exit is unlikely to be a major source of bias in our sample.

4.3 Generating Comparison Groups

To assess the effects of displacement, we construct two comparison groups that provide alternative counterfactuals for displaced workers. Rather than pooling all non-displaced workers within the same MSA, as is common in the literature (Lachowska et al., 2020; Schmieder et al., 2023), we distinguish between workers within and outside the mass layoff establishment.

Specifically, we compare displaced workers to: (1) non-displaced workers employed at the same establishment at the time of the mass layoff (Comparison Group A), and (2) non-displaced workers employed at other establishments in the same MSA (Comparison Group B). Comparison Group B is constructed as a

random sample of workers who do not experience a mass layoff. All workers in both comparison groups are required to satisfy the same baseline restrictions as displaced workers.

Each comparison group has distinct advantages and limitations. Comparison Group A controls for selection into the establishment, as displaced and non-displaced workers were employed at the same firm prior to the layoff. However, the remaining workers may differ systematically from those who are displaced, and their earnings may also be affected by the mass layoff event. As a result, comparisons based on this group may understate the full impact of displacement.

In contrast, Comparison Group B consists of workers in the same labor market who are not directly affected by the mass layoff event. This group provides a benchmark that is less likely to be influenced by establishment-level shocks, but may differ from displaced workers in both observable and unobservable ways.

In our empirical analysis, we report overall results using both comparison groups, but focus the majority of the analysis on Comparison Group B, as this group provides the most compelling counterfactual of how a worker may have fared in the absence of the layoff event. Comparing outcomes across these groups provides insight into the role of establishment-specific shocks and broader labor market conditions. In addition, our earnings specifications include worker fixed effects, which help account for time-invariant differences across workers.

5 Summary Statistics

We begin by presenting descriptive statistics for our sample of 162,000 workers displaced by mass layoffs, along with the two comparison groups. Table 1 summarizes worker characteristics at baseline, including demographics, prior earnings, tenure, and industry. Table 2 provides an overview of jobless duration and re-employment outcomes over the five years following displacement.

Table 1 describes the full sample of 389,000 workers, including workers at mass layoff establishments (columns 1–3) and workers at non-mass layoff establishments (Comparison Group B, column 4). Workers at mass layoff establishments tend to be somewhat younger, have lower prior earnings, and shorter tenure, and are more likely to be male. They are also more concentrated in construction and manufacturing and in smaller establishments relative to workers in non-mass layoff establishments.

Columns 2 and 3 of Table 1 further distinguish between displaced workers and non-separated workers at mass layoff establishments (Comparison Group A). Comparison Group A is broadly similar to Comparison Group B, with only modest differences in gender composition, industry, and establishment size. In contrast, displaced workers differ from both comparison groups: they are younger, have lower prior earnings, and shorter tenure. These differences are consistent with displacement not being randomly assigned within

Table 1: Characteristics of Workers at Mass Layoff and Non-Mass Layoff Establishments

	Mass Layoff Establishments			Non-Mass Layoff
	All Workers (1)	Displaced (2)	Non-Displaced (3)	Non-Displaced (4)
<i>Age</i>				
25–29	0.112	0.130	0.099	0.103
30–34	0.139	0.150	0.132	0.135
35–39	0.162	0.163	0.161	0.159
40–44	0.178	0.173	0.181	0.174
45–49	0.196	0.185	0.203	0.200
50–55	0.214	0.199	0.225	0.228
<i>Earnings</i>				
\$15,000–\$29,999	0.252	0.312	0.209	0.230
\$30,000–\$44,999	0.305	0.297	0.311	0.288
\$45,000–\$59,999	0.192	0.163	0.213	0.213
\$60,000–\$74,999	0.111	0.092	0.124	0.123
\$75,000–\$89,999	0.059	0.054	0.062	0.061
\$90,000+	0.082	0.081	0.082	0.085
<i>Tenure</i>				
4 Quarters	0.242	0.292	0.206	0.176
5–8 Quarters	0.305	0.336	0.283	0.262
9–16 Quarters	0.162	0.143	0.175	0.192
16+ Quarters	0.292	0.230	0.337	0.371
<i>Race/Ethnicity</i>				
Non-Hispanic White	0.844	0.811	0.867	0.876
Black	0.088	0.111	0.071	0.075
Non-Black Hispanic	0.041	0.046	0.037	0.025
Other	0.028	0.032	0.025	0.023
<i>Gender</i>				
Male	0.641	0.597	0.673	0.483
Female	0.359	0.403	0.327	0.518
<i>Industry</i>				
Construction/Utilities	0.092	0.062	0.114	0.030
Manufacturing	0.422	0.349	0.474	0.180
Trade/Transport	0.139	0.186	0.105	0.137
Finance/Professional	0.209	0.233	0.193	0.196
Education/Health	0.092	0.127	0.067	0.335
Leisure/Hospitality	0.022	0.030	0.016	0.024
Other	0.024	0.013	0.031	0.098
<i>Establishment Size</i>				
50–500 Employees	0.704	0.698	0.709	0.579
501–2000 Employees	0.204	0.192	0.213	0.255
2000+ Employees	0.092	0.110	0.079	0.167
Number of Observations	389,000	162,000	227,000	238,000

Notes: Calculations use LEHD data. Characteristics are measured at baseline. Earnings are expressed in 2000 dollars. The sample includes displaced workers, non-displaced workers at mass layoff establishments, and non-displaced workers at non-mass layoff establishments for whom stable employment outcomes can be observed.

Table 2: Transitions to a Stable Job after Displacement

	Share
<i>Employment Status (20 Quarters after Displacement)</i>	
Still without stable job	9.2%
Re-employed, same MSA	67.2%
Re-employed, different MSA	23.6%
<i>Joblessness Duration</i>	
<1 Quarter	36.1%
1 Quarter	25.8%
2–3 Quarters	8.9%
4–7 Quarters	11.1%
8+ Quarters	18.0%
Number of Observations	162,000

Notes: Calculations use LEHD data. The sample includes displaced workers for whom stable re-employment outcomes can be observed following a mass layoff.

establishments.

A potential concern is that workers may anticipate mass layoffs and exit prior to the event. If so, the remaining workers could differ systematically from those who would be observed under unanticipated displacement. To assess this possibility, we compare workers who leave in the four quarters preceding the mass layoff (“early leavers”) to those who leave in the preceding four quarters (“very early leavers”). Appendix C, Table C1 shows that these groups are very similar across observable characteristics, suggesting that anticipatory exit is unlikely to be a major source of bias.

Table 2 summarizes re-employment outcomes for displaced workers. Approximately two-thirds of displaced workers find stable re-employment in their origin MSA. However, the distribution of jobless duration exhibits a substantial right tail, with 9.2 percent of workers remaining non-employed five years after displacement. While most workers are re-employed locally, a sizable share—23.6 percent—find employment in a different MSA, highlighting the potential importance of geographic mobility.

6 Empirical Framework

This section outlines our empirical approach to examining how local labor market conditions and outside options shape post-displacement outcomes. Guided by the conceptual framework, we focus on two key dimensions of adjustment: (i) earnings dynamics following job loss and (ii) the duration and location of re-employment. Together, these approaches allow us to assess both the consequences of displacement and the channels through which local and outside labor market conditions affect worker outcomes.

6.1 Earnings Dynamics after Displacement

We begin by examining the evolution of earnings following displacement. Our analysis includes 143,000 displaced workers for whom we observe complete earnings histories, along with the two comparison groups described above: non-separated workers at mass layoff establishments (Comparison Group A) and workers at non-mass layoff establishments in the same labor market (Comparison Group B).

We estimate an event-study specification of the form:

$$EARN_{imst} = \alpha_i + \gamma_{st} + \sum_{k=-4}^{23} \delta_k D_{it}^k + \epsilon_{imst} \quad (2)$$

where $EARN_{imst}$ denotes real earnings of worker i in MSA m , state s , and quarter t . The indicators D_{it}^k capture event time relative to displacement, allowing earnings to evolve flexibly before and after the mass layoff event. The specification includes worker fixed effects, α_i , and state-by-quarter fixed effects, γ_{st} , so identification is based on within-worker changes in earnings over time. One pre-displacement period is omitted as the reference category.

For displaced workers, event time is defined relative to the quarter of job loss. To align the timing of earnings responses across workers displaced at different points within the layoff window, we normalize event time relative to the period immediately preceding the largest earnings disruption. For workers in the comparison groups, we assign event time based on the timing of the mass layoff event in their establishment (Comparison Group A) or draw a comparison quarter from the empirical distribution of layoff timing (Comparison Group B).

To assess how labor market conditions shape earnings trajectories, we interact event-time indicators with measures of local and outside labor market conditions at the time of displacement, including unemployment rates (UNEMP), the Job Opportunities Index (JOI), and housing costs (HCI). We also allow earnings paths to vary by industry and by whether re-employment occurs in the origin or an outside MSA.

6.2 Jobless Duration and Mobility

We next examine the duration of non-employment and the location of re-employment following displacement. This analysis uses the full sample of displaced workers, including those for whom complete earnings histories are not observed.

We define re-employment as the first occurrence of four consecutive quarters of positive earnings, capturing transitions into stable employment. This definition allows us to include workers who move to states outside our earnings sample.

To model re-employment outcomes, we estimate a competing risks framework in which workers transition from non-employment to either (i) employment in the origin MSA or (ii) employment in an outside MSA. Let Y_{id} denote the outcome for worker i at duration d , taking values corresponding to non-employment ($k = 1$), employment in the origin MSA ($k = 2$), and employment in an outside MSA ($k = 3$). Outcomes are determined by a standard discrete choice framework:

$$Y_{id} = \begin{cases} 1 & \text{if } U_{id}^1 = \max\{U_{id}^1, U_{id}^2, U_{id}^3\} \\ 2 & \text{if } U_{id}^2 = \max\{U_{id}^1, U_{id}^2, U_{id}^3\} \\ 3 & \text{if } U_{id}^3 = \max\{U_{id}^1, U_{id}^2, U_{id}^3\} \end{cases} \quad (3)$$

We specify a multinomial logit model in which transition probabilities depend on worker characteristics, duration dependence, and labor market conditions in both the origin and outside option MSAs at the time of displacement:

$$U_{id}^k = \alpha_k + X_i \beta_k + MSA_i^O \gamma_k + MSA_i^N \delta_k + \pi_{st} + h(d; \lambda_k) + \epsilon_{id}^k \quad (4)$$

where U_{id}^k denotes the latent utility associated with transition k at duration d . The vector X_i includes worker characteristics, while MSA_i^O and MSA_i^N capture labor market conditions in the origin and outside option MSAs at the time of displacement, including UNEMP, JOI, and HCI. The function $h(d; \lambda_k)$ captures duration dependence, and π_{st} denotes state-by-quarter fixed effects. We assume that the error terms ϵ_{id}^k are independently and identically distributed with a Type I extreme value distribution, implying a multinomial logit specification for transition probabilities.

This framework allows us to separately estimate how local and outside labor market conditions affect both the likelihood of re-employment and the location in which re-employment occurs, consistent with the conceptual framework's emphasis on outside options and spatial frictions.

Consistent with the conceptual framework, we expect stronger labor market conditions—captured by lower unemployment rates and higher JOI—to increase the likelihood of re-employment and shorten jobless durations. Improvements in outside labor markets should similarly increase transitions to employment in those locations. In contrast, higher housing costs are expected to act as a barrier to mobility, reducing transitions to outside MSAs.

Because the competing risks specification does not include worker fixed effects, we assess robustness to unobserved heterogeneity using alternative bivariate logit models with individual heterogeneity modeled using both a two-point distribution (Heckman and Singer, 1982) and a Gamma distribution.

7 Results

This section presents the main empirical results. We first examine earnings dynamics following displacement, and then turn to the jobless duration and mobility decisions. Together, these results allow us to assess both the consequences of displacement and the role of local labor market conditions and spatial frictions in shaping worker adjustment.

7.1 Earnings Results

We begin by examining how earnings evolve following displacement and how these trajectories vary with local labor market conditions. In the conceptual framework, differences in local opportunities shape both the speed of re-employment and the quality of jobs obtained. The earnings results therefore provide a reduced-form summary of how these forces translate into long-run outcomes.

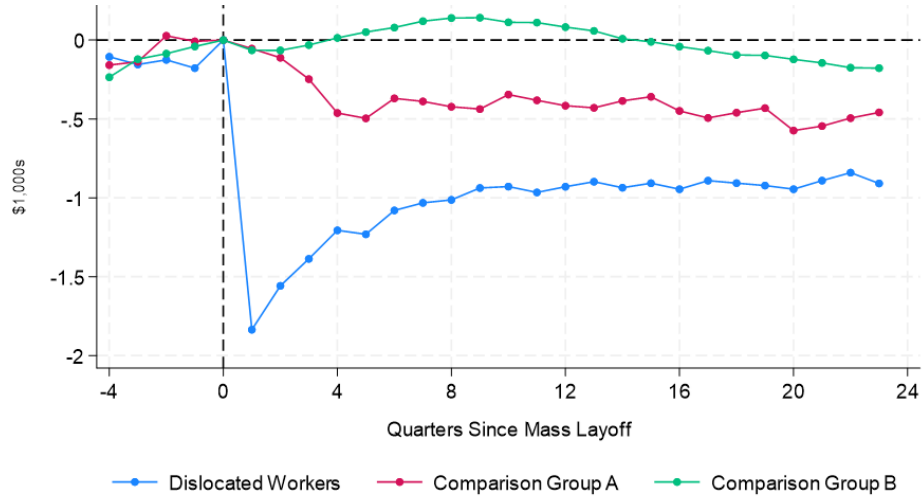
Figure 2, panel (a) shows the change in real quarterly earnings for three groups: workers displaced following a mass layoff (Dislocated Workers), non-separated workers at mass layoff establishments (Comparison Group A), and workers at non-mass layoff establishments (Comparison Group B). Figure 2, panel (b) presents the corresponding treatment effects, measured as the percent difference in earnings relative to the comparison groups.

Pre-displacement earnings paths are similar across all three groups, supporting the use of workers at non-mass layoff establishments as a valid comparison group. In contrast, displaced workers experience a sharp decline in earnings in the first quarter following job loss—nearly 30 percent on average. Earnings recover gradually over time, but even after five years remain 7.2 and 11.4 percent below Comparison Groups A and B, respectively. These results are robust to alternative specifications, including propensity score weighting (see Appendix Figure D1).

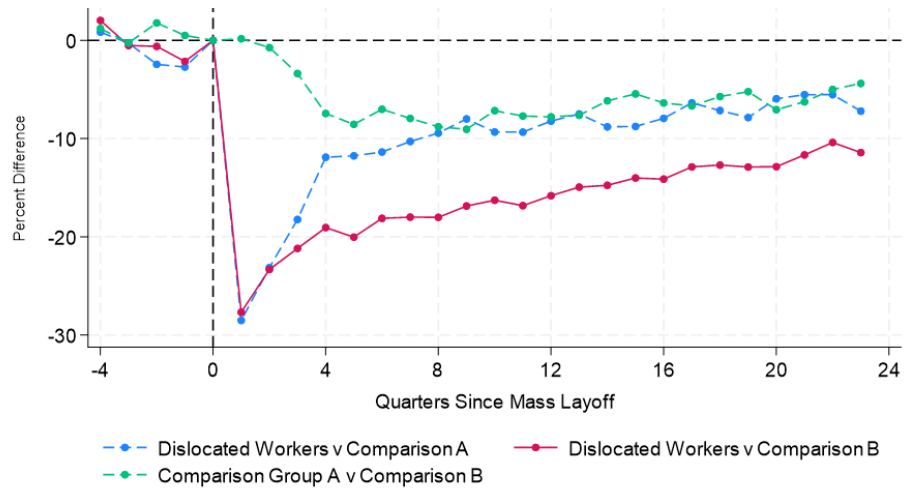
This average masks substantial heterogeneity in worker experiences, which is central to our second contribution. While some workers return quickly to full-time employment, others experience prolonged joblessness (as shown in Table 2). Figure 3 highlights heterogeneity across industries. Earnings losses are largest in manufacturing, where displaced workers experience a 43 percent decline in earnings immediately following displacement and remain 18 percent below baseline after five years. In contrast, workers in accommodation and food services experience much smaller losses. These patterns are consistent with the broader literature documenting large and persistent earnings losses for workers displaced from manufacturing during the Great Recession (Davis and von Wachter, 2011; Charles et al., 2019). They also reflect differences in the transferability of skills and the availability of comparable job opportunities across sectors, reinforcing the role of

Figure 2: Earnings, Relative to Mass Layoff

(a) Quarterly Earnings by Worker Group

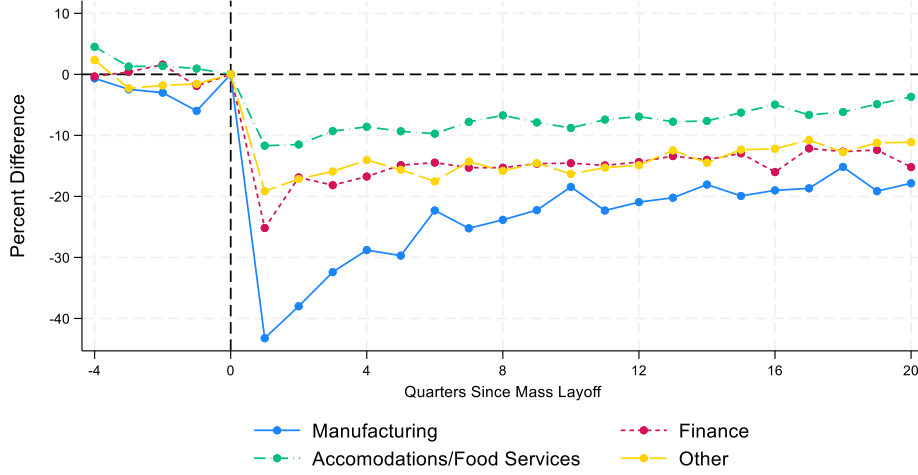


(b) Percent Differences in Earnings Relative to Comparison Groups



Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to mass layoff. Comparison Group A represents workers who were not separated from mass layoff establishments. Comparison Group B represents workers who were not employed in mass layoff establishments.

Figure 3: Percent Difference in Earnings, Relative to Comparison Group by Industry



Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to Mass Layoff. Differences are relative to Comparison Group B in the same industry. Comparison Group B represents workers who were not employed in mass layoff establishments.

local opportunity structures emphasized in our analysis.

We next turn to our key measures of local labor market conditions: unemployment rates (UNEMP), the Job Opportunities Index (JOI), and housing costs (HCI). Figures 4-6 display earnings trajectories for displaced workers across different points in the distribution of these measures in both the origin and outside option MSAs. These paths are evaluated at the 10th, 25th, 50th, 75th, and 90th percentiles of UNEMP and the JOI and HCI indices.

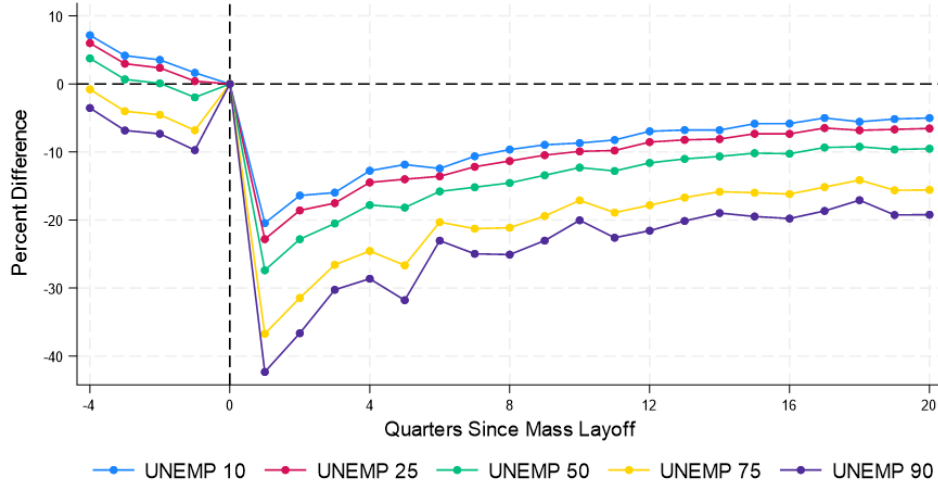
The most striking pattern emerges for the JOI. Workers displaced in high-JOI labor markets experience substantially smaller earnings losses, and those in the top decile fully recover their pre-displacement earnings within five years. In contrast, workers in low-JOI markets experience persistent earnings losses. This highlights the importance of local opportunity structures, as captured by the JOI, in shaping long-run adjustment.

We also find that lower unemployment rates in the origin MSA are associated with faster and more complete earnings recovery, though the magnitude of these effects is somewhat smaller than for the JOI. In contrast, variation in housing costs has little impact on earnings trajectories, either in the origin or outside option MSAs.

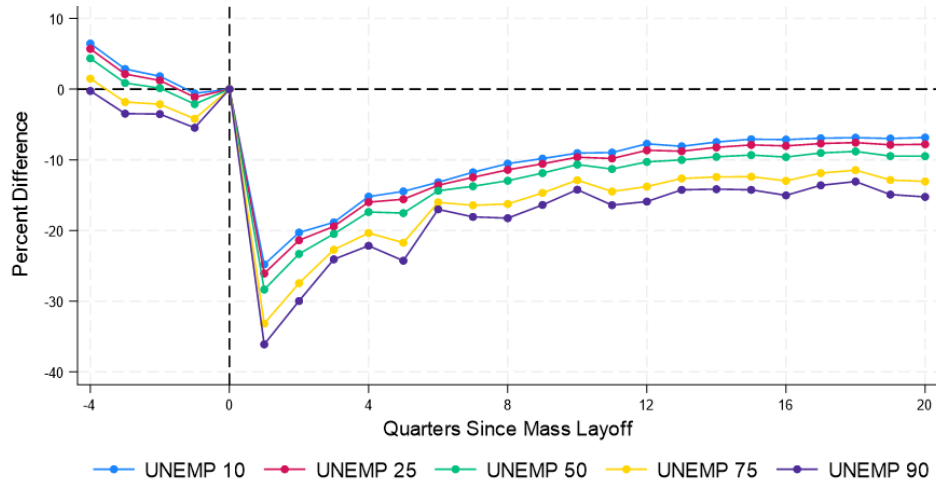
There is also some evidence that labor market conditions in outside option MSAs matter, though the effects are limited. In particular, lower unemployment rates in outside option markets are associated with

Figure 4: Percent Differences in Earnings, by Unemployment Rate Percentiles

(a) Interacted with Local Unemployment Rate Percentiles



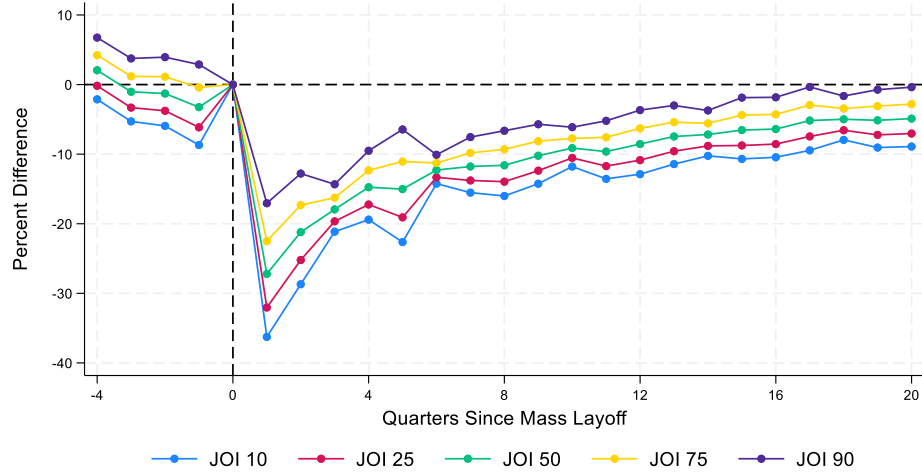
(b) Interacted with Outside Option Unemployment Rate Percentiles



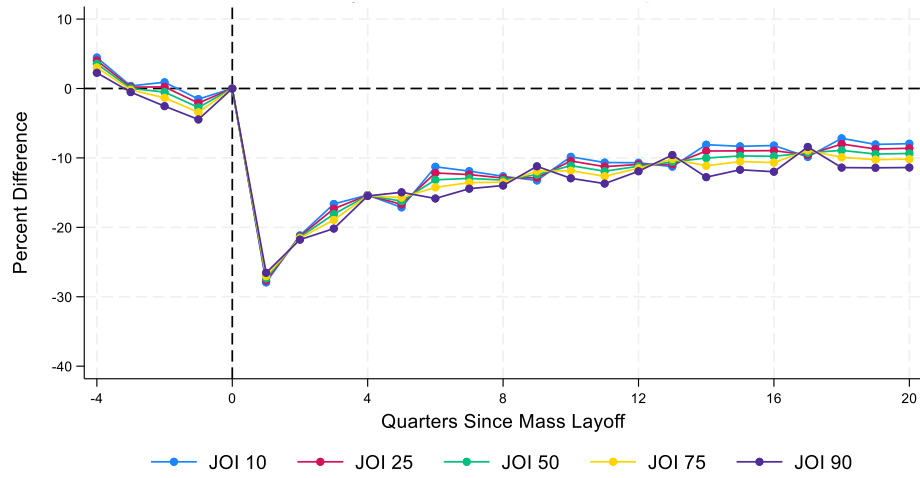
Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to mass layoff and relative to Comparison Group B. Comparison Group B represents workers who were not employed in mass layoff establishments. The underlying regression controls for JOI and HCI.

Figure 5: Percent Differences in Earnings, by JOI Percentiles

(a) Interacted with Local JOI Percentiles



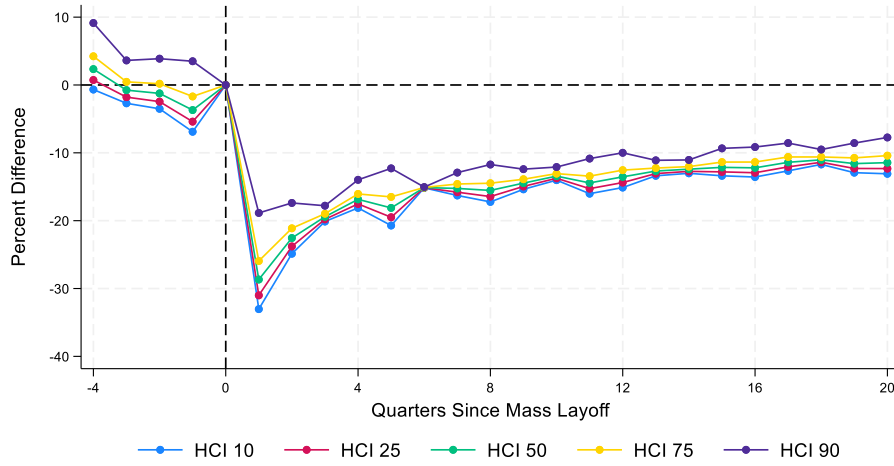
(b) Interacted with Outside Option JOI Percentiles



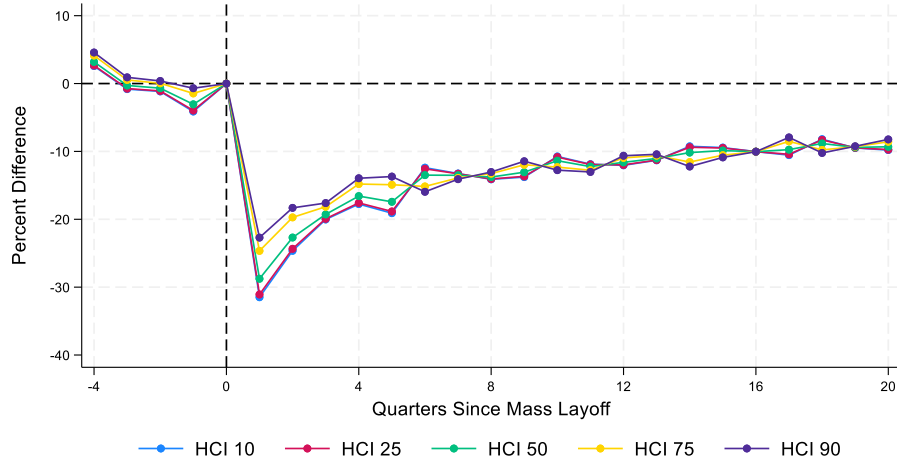
Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to mass layoff and relative to Comparison Group B. Comparison Group B represents workers who were not employed in mass layoff establishments. The underlying regression controls for UNEMP and HCI.

Figure 6: Percent Differences in Earnings, by HCI Percentiles

(a) Interacted with Local HCI Percentiles



(b) Interacted with Outside Option HCI Percentiles



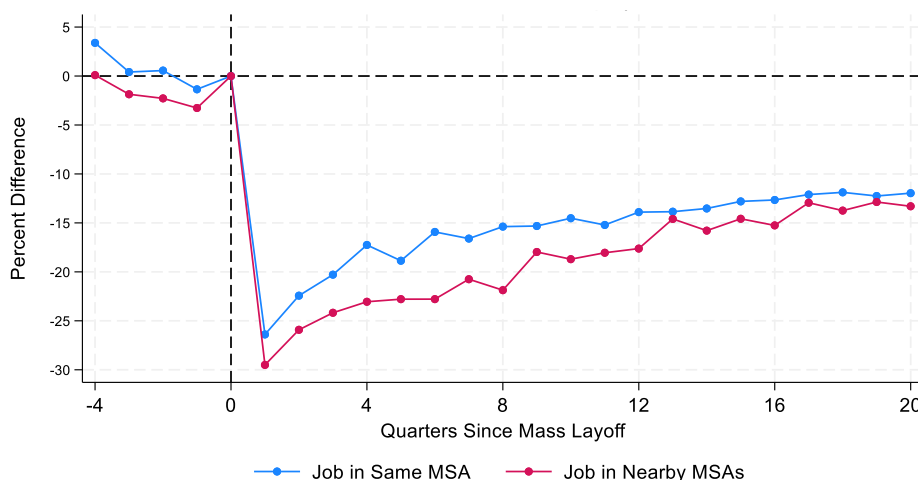
Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to mass layoff and relative to Comparison Group B. Comparison Group B represents workers who were not employed in mass layoff establishments. The underlying regression controls for unemployment rate and JOI.

modest improvements in earnings outcomes, while we find little evidence that variation in the JOI or housing costs in outside MSAs affects earnings trajectories.

These results highlight substantial heterogeneity in recovery across labor markets. A natural question is whether workers are able to mitigate these differences through geographic mobility. In the conceptual framework, workers may respond to poor local opportunities by relocating to higher-opportunity regions. We next examine whether mobility serves this role.

Figure 7 compares earnings trajectories for displaced workers who are re-employed in their origin MSA versus those who move to a different MSA. While movers experience slightly larger initial earnings losses, they eventually converge to similar earnings paths as non-movers. This pattern suggests that mobility does not systematically lead to improved long-run earnings outcomes. Although these comparisons are subject to selection, they are consistent with the presence of spatial frictions that limit the extent to which workers can benefit from relocating to higher-opportunity regions.

Figure 7: Percent Difference in Earnings, Based on Re-employment Location



Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to Mass Layoff. Differences are relative to Comparison Group B. Comparison Group B represents workers who were not employed in mass layoff establishments.

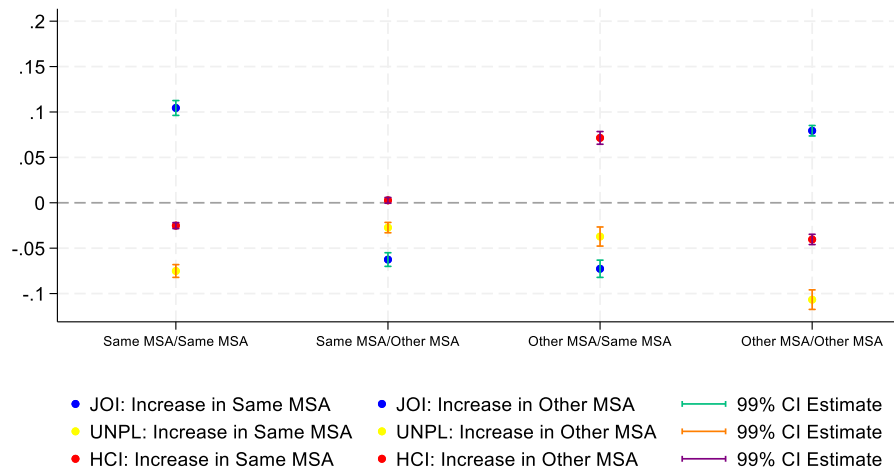
Taken together, these results highlight two key findings. First, there is substantial heterogeneity in earnings recovery across labor markets, driven by differences in local job opportunities. Second, mobility does not appear to equalize these differences, suggesting limits to the role of geographic adjustment in mitigating the costs of displacement.

7.2 Jobless Duration and Mobility Results

To further evaluate these patterns within the framework of worker location choices, we turn to the jobless duration and mobility results. In the conceptual framework, displaced workers choose between remaining non-employed, accepting a job in the origin MSA, or moving to an outside labor market. The competing risks framework allows us to assess how local and outside labor market conditions affect both the timing and location of re-employment.

Figure 8 summarizes the estimated effects of local and outside labor market conditions on re-employment outcomes. For completeness, the full set of coefficient estimates underlying these results is reported in Appendix Table E1. The model produces separate parameter estimates for transitions into employment in the origin MSA and in outside MSAs, allowing us to distinguish between local and nonlocal adjustment.

Figure 8: Standardized Coefficients for Impact of UNEMP, JOI, HCI on Exit from Joblessness



Notes: Values in this table are results from estimating the competing risks model of jobless duration and mobility (equations 3 and 4) using the LEHD data. Indicators on the X axis refer to $-(\text{the source of the change})/(\text{the location of the impact})$. The capped lines around the point estimates are the 99 percent confidence interval estimates.

We begin with the direct effects of labor market conditions in the origin MSA. A one standard deviation increase in the unemployment rate reduces the likelihood of re-employment in the origin MSA by 0.08 standard deviations, while a comparable increase in the JOI increases it by 0.10 standard deviations. These effects are economically large and approximately symmetric in magnitude, highlighting that the JOI captures an additional dimension of labor market opportunity beyond unemployment. In contrast, the effect of housing costs is smaller, though higher housing costs reduce the likelihood of re-employment.

A similar pattern emerges for labor market conditions in outside MSAs. Stronger conditions in outside

markets—lower unemployment and higher JOI—increase the likelihood of re-employment in those locations. These results reflect a “pull” effect, whereby better opportunities attract workers into those labor markets.

We also examine cross-effects, defined as the impact of conditions in one location on transitions to another. In general, improvements in labor market conditions in one location reduce transitions to other locations. In particular, higher JOI in outside MSAs reduces the likelihood of re-employment in the origin MSA, consistent with workers extending their job search to access better opportunities elsewhere. Housing costs exhibit the opposite pattern, with higher housing costs in outside MSAs reducing mobility into those markets, consistent with housing acting as a barrier to relocation.

These findings help reconcile the weaker role of outside labor market conditions in the earnings results above. While we found only limited effects of outside options on earnings trajectories, the results here show that outside labor market conditions play a significant role in determining the timing and location of re-employment. In this sense, outside options primarily operate along the extensive margin—whether and where workers find jobs—rather than the intensive margin of earnings conditional on re-employment. Combined with spatial frictions that limit mobility, this helps explain why outside opportunities have a muted impact on observed earnings outcomes.

8 Conclusion

This paper examines how local and outside option labor market conditions shape the adjustment of workers displaced during the Great Recession. Using matched employer-employee data from the LEHD combined with measures of local labor market conditions, we study both earnings dynamics and the duration and location of re-employment. Guided by a conceptual framework in which workers choose between remaining non-employed, accepting local employment, or relocating to another labor market, our empirical approach allows us to link post-displacement outcomes to both local opportunities and outside options.

We make three main contributions. First, we introduce a new measure of local labor market opportunity, the Job Opportunities Index (JOI), which captures both the likelihood of finding a job and the expected earnings associated with new employment. Second, we document substantial heterogeneity in post-displacement outcomes across labor markets, showing that workers in higher-opportunity areas experience significantly smaller earnings losses. Third, we provide new evidence on the role of spatial frictions, showing that geographic mobility does not fully offset differences in local labor market conditions.

Consistent with the existing literature, we find large and persistent earnings losses following displacement, averaging approximately 12 percent five years after job loss. These losses are particularly pronounced in manufacturing, reflecting the sector’s central role in the Great Recession and the limited transferability of

sector-specific skills. We also document previously underexplored losses among non-separated workers at mass layoff establishments, suggesting that the effects of displacement extend beyond those directly laid off.

A central finding of the paper is the importance of local labor market conditions in shaping recovery. Workers displaced in high-opportunity labor markets, as measured by the JOI, experience substantially smaller earnings losses, and those in the highest-opportunity markets fully recover their pre-displacement earnings within five years. In contrast, workers in low-opportunity markets experience persistent losses. These results highlight the role of local opportunity structures in determining long-run outcomes.

To further evaluate the implications of the conceptual framework, we estimate a competing risks model of jobless duration and mobility. We find that both local and outside labor market conditions significantly affect the likelihood and location of re-employment. Stronger labor markets increase the probability of re-employment both locally and in nearby regions, reflecting a “pull” effect of job opportunities. At the same time, housing costs act as a barrier to mobility, reducing transitions into higher-cost labor markets.

Taken together, these findings suggest that workers respond to differences in labor market conditions but do not fully adjust to them. Mobility does not eliminate disparities in outcomes across locations, consistent with the presence of spatial frictions emphasized in the conceptual framework.

These results have several policy implications. First, the strong role of local labor market conditions suggests that policies supporting displaced workers may be more effective when tailored to local economic conditions. Programs such as National Dislocated Worker Grants could incorporate local labor market indicators when allocating resources. Second, our findings support ongoing interest in place-based policies that improve local job opportunities, particularly in regions experiencing persistent economic shocks. Finally, the role of housing costs in limiting mobility suggests that policies aimed at reducing barriers to relocation—such as housing assistance—may improve access to higher-opportunity labor markets.

More broadly, our findings highlight the importance of considering both local opportunities and mobility constraints when evaluating the adjustment of workers to economic shocks. While displaced workers respond to differences in labor market conditions, spatial frictions limit their ability to fully access better opportunities, contributing to persistent inequality in post-displacement outcomes.

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Appendix A Construction of the Job Opportunities Index (JOI)

This appendix provides additional detail on the construction of the Job Opportunities Index (JOI) and the alternative specifications we consider.

A.1 Data Sources and Basic Construction

We construct the JOI using publicly available data from the Longitudinal Employer-Household Dynamics (LEHD) Job-to-Job Flows (J2J) dataset. The J2J data provide MSA-level information on hires, separations, total employment, and earnings for 383 metropolitan statistical areas (MSAs). These measures are available at quarterly frequency and can be disaggregated by industry (20 NAICS sectors) and by worker education level (three categories).

These data allow us to construct measures of job opportunities that vary across both labor markets and worker skill groups. In the main analysis, we define the JOI separately by MSA, education group, and quarter, and assign workers the JOI corresponding to their education group.

A.2 Definition of the JOI

The JOI is designed to capture the expected employment value of a labor market by combining information on both the likelihood of obtaining a job and the expected earnings conditional on employment. Formally, for MSA m , education group e , and time t , the JOI is defined as:

$$JOI_{met} = \bar{w}_{met} \times \lambda_{met} \quad (5)$$

where $\bar{w}_{met}$ is the average earnings of job-to-job switchers into MSA m and λ_{met} is a measure of hiring intensity.

Expected earnings are measured as the average quarterly earnings of workers who switch jobs into the MSA, capturing the wage component of available opportunities. The hiring intensity term is intended to capture the likelihood of obtaining employment in that labor market.

A.3 Alternative Measures of Hiring Intensity

We consider four alternative measures of hiring intensity:

1. Total hires in the MSA
2. Net hires, defined as hires minus separations

3. The hiring rate, defined as hires divided by total employment
4. The net hiring rate, defined as net hires divided by total employment

These measures capture different aspects of labor market dynamics. Total hires reflect the volume of job creation, while net hires account for the extent to which new jobs are offset by separations. Scaling these measures by total employment yields hiring rates that are comparable across labor markets of different sizes.

The hiring rate provides a measure of job creation relative to the size of the labor market, while the net hiring rate additionally incorporates separations and therefore reflects the stability of job creation. In particular, higher net hiring rates indicate that new jobs are less likely to be offset by separations, suggesting more persistent employment opportunities.

A.4 Selection of the Baseline Measure

While each of these measures captures a relevant dimension of labor market conditions, they differ in their empirical performance. To guide our choice, we examine the relationship between these alternative measures and earnings outcomes for job switchers.

Specifically, we compute the correlation between each measure of hiring intensity and the average change in earnings for workers who switch jobs into a given MSA. This outcome captures an important dimension of job quality and is directly related to the earnings component of the JOI.

Table A1 presents summary statistics for hires, separations, employment, hiring rates, net hiring rates, and earnings changes across MSAs from 2001 to 2017. Table A2 reports results from regressions of earnings changes on the alternative measures of hiring intensity. We standardize all variables, so the estimated coefficients can be interpreted as standardized relationships.

Table A1: Summary Statistics for J2J Measures Across MSAs

Variable	Mean	Std. Dev.	Min	Max
Hires (thousands)	30.622	65.050	1.047	876.804
Separations (thousands)	29.485	62.259	0.971	858.367
Net Hire (thousands)	1.138	4.391	-45.627	65.034
Employed (thousands)	276.520	619.426	7.284	8760.782
Hire Rate	0.116	0.020	0.058	0.236
Net Hire Rate	0.004	0.007	-0.138	0.051
Average % Change in Earnings for J2J Workers	7.311	4.206	-30.777	50.757

Notes: Summary statistics are calculated across MSAs using J2J data from 2001.

Across specifications, the hiring rate and net hiring rate exhibit stronger and more robust relationships with earnings changes than measures based on total hires or net hires. In particular, the net hiring rate is

Table A2: Regression Results for Alternative JOI Components

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Net Hire Rate	0.315*** (0.021)	0.221*** (0.023)							0.173*** (0.023)
Hire Rate			-0.012 (0.014)	0.430*** (0.032)					0.241*** (0.040)
Net Hires					0.091*** (0.013)	0.042*** (0.013)			-0.048*** (0.015)
Hires							-0.046*** (0.010)	0.333*** (0.061)	0.220*** (0.070)
MSA Effects	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Year Effects	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Obs.	6879	6879	6879	6879	6879	6879	6879	6879	6879
R-squared	0.099	0.530	0.000	0.525	0.008	0.501	0.002	0.501	0.538
Mean of Dep. Var	0.073	0.073	0.073	0.073	0.073	0.073	0.073	0.073	0.073

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable in all columns is the standardized average change in earnings for J2J workers switching to a job in the destination MSA. All explanatory variables are standardized, so coefficient estimates can be interpreted as the number of standard deviations in the dependent variable associated with a one standard deviation change in the explanatory variable.

consistently positively associated with earnings gains for job switchers, both in simple correlations and in specifications that include MSA and time fixed effects.

Based on this evidence, we use the net hiring rate as our baseline measure of hiring intensity in constructing the JOI.

A.5 Interpretation

The resulting JOI combines information on both the quantity and quality of job opportunities in a labor market. Higher values of the JOI reflect labor markets with both greater hiring intensity and higher expected earnings, and therefore greater overall opportunity for displaced workers.

Because the JOI is constructed separately by education group, it captures differences in opportunities across skill levels as well as across locations. In the empirical analysis, we assign workers the JOI corresponding to their education group, allowing us to examine how variation in local labor market conditions affects outcomes within skill groups.

Appendix B Assignment of Workers to Establishments

This appendix describes the procedure used to assign workers to establishments within multi-establishment firms. This step is necessary because the LEHD provides worker-level data at the firm level, while our analysis is conducted at the establishment level.

For multi-establishment firms, we rely on a probabilistic assignment method developed by the Census Bureau ([Abowd et al., 2009](#)). This method uses information from Minnesota, where unemployment insurance records are available at both the firm and establishment levels, to infer likely worker-establishment matches in other states. The assignment is based on a linear spline in the distance between the worker’s residence and the location of each establishment within the firm.

For each worker, the Census provides 10 imputed establishment assignments. When establishments are geographically dispersed, these imputations tend to coincide, resulting in a high degree of certainty about the worker’s assigned location. In contrast, for firms with multiple nearby establishments—particularly in dense urban areas—workers may be assigned to multiple plausible establishments.

To minimize measurement error, we adopt a conservative assignment rule. We assign a worker to an establishment only if at least seven of the ten imputations correspond to the same establishment. Workers who do not meet this criterion are excluded from the analysis. Because our sample is restricted to establishments with at least 50 workers, ambiguity due to many small establishments is limited. This approach reduces the risk of misclassification while preserving a large share of the sample.

Appendix C Early Leavers

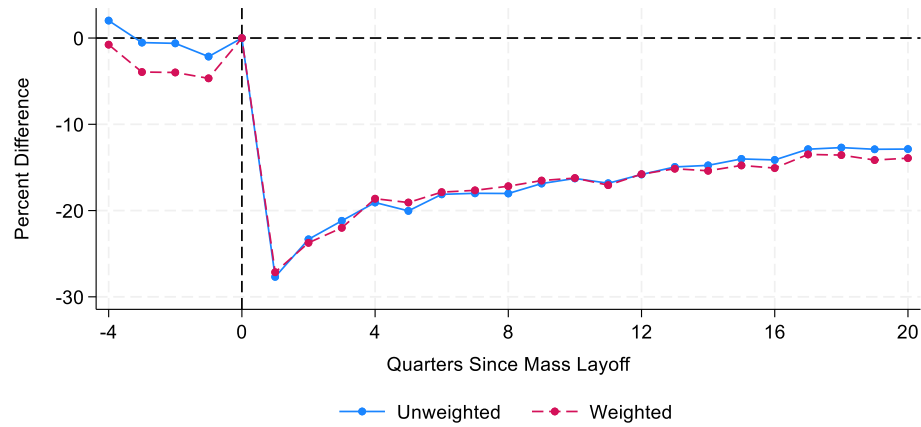
Table C1: Characteristics of Workers Leaving before Mass Layoff Event

	All Workers	Displaced Workers	Early Leavers (1–4Q)	Very Early Leavers (5–8Q)
	(1)	(2)	(3)	(4)
<i>Age</i>				
25–29	0.112	0.130	0.193	0.193
30–34	0.139	0.150	0.164	0.165
35–39	0.162	0.163	0.166	0.167
40–44	0.178	0.173	0.161	0.163
45–49	0.196	0.185	0.159	0.158
50–55	0.214	0.199	0.158	0.154
<i>Earnings</i>				
\$15,000–\$29,999	0.252	0.312	0.379	0.400
\$30,000–\$44,999	0.305	0.297	0.284	0.278
\$45,000–\$59,999	0.192	0.163	0.151	0.153
\$60,000–\$74,999	0.111	0.092	0.084	0.081
\$75,000–\$89,999	0.059	0.054	0.043	0.039
\$90,000+	0.082	0.081	0.060	0.050
<i>Tenure</i>				
4 Quarters	0.242	0.292	0.480	0.507
5–8 Quarters	0.305	0.336	0.287	0.251
9–16 Quarters	0.162	0.143	0.095	0.118
16+ Quarters	0.292	0.230	0.139	0.124
<i>Race/Ethnicity</i>				
Non-Hispanic White	0.844	0.811	0.792	0.793
Black	0.088	0.111	0.127	0.127
Non-Black Hispanic	0.041	0.046	0.051	0.050
Other	0.028	0.032	0.030	0.030
<i>Gender</i>				
Male	0.641	0.597	0.604	0.599
Female	0.359	0.403	0.397	0.401
<i>Industry</i>				
Construction/Utilities	0.092	0.062	0.091	0.093
Manufacturing	0.422	0.349	0.301	0.279
Trade/Transport	0.139	0.186	0.134	0.144
Finance/Professional	0.209	0.233	0.290	0.295
Education/Health	0.092	0.127	0.120	0.143
Leisure/Hospitality	0.022	0.030	0.046	0.032
Other	0.024	0.013	0.017	0.015
Number of Observations	389,000	162,000	108,000	60,500

Notes: Calculations use LEHD data. Earnings are expressed in 2000 dollars. The sample includes displaced workers, early leavers, and very early leavers.

Appendix D Propensity Score Weighted Earnings Results

Figure D1: Percent Difference in Earnings, Unweighted and Weighted Regressions



Notes: Calculations use LEHD data for all establishments in the five sample states. Includes all individuals experiencing mass layoffs for whom we can identify earnings. Differences are relative to earnings in the quarter prior to Mass Layoff and relative to Comparison Group B. Comparison Group B represents workers who were not employed in mass layoff establishments. Weighted results are based on inverse propensity score weights.

Appendix E Duration/Mobility Model Results

Table E1: Probability of Finding New Employment by Geographic Location

	Estimates		Standardized Coefficients	
	Same MSA (1)	Different MSA (2)	Same MSA (3)	Different MSA (4)
Unemployment Rate: Origin MSA	-0.065 (0.003)**	-0.039 (0.004)**	-0.075	-0.027
Unemployment Rate: Outside Option MSA	-0.039 (0.006)**	-0.183 (0.009)**	-0.037	-0.107
Job Opportunity Index: Origin MSA	0.005 (0.0002)**	-0.005 (0.0003)**	0.104	-0.063
Job Opportunity Index: Outside Option MSA	-0.003 (0.0002)**	0.005 (0.0002)**	-0.073	0.079
Housing Cost: Origin MSA	-0.003 (0.0002)**	0.001 (0.0003)	-0.025	0.003
Housing Cost: Outside Option MSA	0.006 (0.0003)**	-0.006 (0.0004)**	0.072	-0.040
Observations	813,000	813,000		
Individuals	162,000	162,000		

Notes: Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Calculations from LEHD data. Includes all individuals experiencing mass layoffs for whom we can identify stable employment. Estimates are presented in columns (1) and (2), and standardized coefficients in columns (3) and (4).

Table E2: Alternative Hazard Specifications

	Logit		Heckman-Singer		Gamma	
	Same MSA (1)	Diff MSA (2)	Same MSA (1)	Diff MSA (2)	Same MSA (3)	Diff MSA (3)
UNEMP: Orig. MSA	-0.074	-0.029	-0.083	-0.020	-0.091	-0.021
UNEMP: O.O. MSA	-0.040	-0.110	-0.018	-0.158	-0.023	-0.153
JOI: Orig. MSA	0.104	-0.060	0.070	-0.033	0.077	-0.032
JOI: O.O. MSA	-0.073	0.078	-0.027	0.034	-0.029	0.035
HCI: Orig. MSA	-0.024	0.003	-0.011	0.003	-0.013	0.001
HCI: O.O. MSA	0.069	-0.039	0.069	-0.047	0.081	-0.046
Observations	813,000	813,000				
Individuals	162,000	162,000				

Notes: Calculations from LEHD data. Includes all individuals experiencing mass layoffs for whom we can identify stable employment. Standardized coefficients are reported.